Tianyi Yu

Contact Information

LACIM
Université du Québec à Montréal
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Education

University of California, San Diego , La Jolla, CA, USA 2020–2024
PhD in Mathematics, Advisor: Brendon Rhoades

University of California, San Diego , La Jolla, CA, USA 2016–2020
Bachelor of Science in Mathematics & Bachelor of Science in Computer Science

Employment

University of Illinois Urbana-Champaign , Urbana, IL, USA Jan. 2027–
J. L. Doob Research Assistant Professor,
Mentor: Alex Yong

Laboratoire d'Algèbre, de Combinatoire et d'Informatique Mathématique ,
Montréal, QC, Canada Oct. 2024–Dec. 2026
CRM–ISM Postdoctoral Fellow ,
Mentors: François Bergeron, Jake Levinson, Alejandro Morales, Hugh Thomas, and Franco Saliola

Research Interests

Algebraic combinatorics and Schubert calculus; combinatorial algebraic geometry; equivariant and K-theoretic Schubert calculus; symmetric functions and crystals.

Preprints

[1] *A positive combinatorial formula for the double Edelman–Greene coefficients* , with Chou, C.
Preprint arXiv:2512.12185 .

Publications and Accepted Papers

[2] *Grothendieck positivity for normal square root crystals* , with Marberg, E. & Tong, K. H.
Advances in Mathematics **501** (2026), 111118.

[3] *Tableau formula for vexillary double Edelman–Greene coefficients* , with Gregory, A. & Hamaker, Z.
Selecta Mathematica (2026).

[4] *Marked bumpless pipedreams and compatible pairs* , with Huang, D. & Shimozono, M.
Combinatorial Theory (accepted).

[5] *Embedding bumpless pipedreams as Bruhat chains* .
International Mathematics Research Notices (2025).

[6] *Grothendieck polynomials of inverse fireworks permutations* , with Chou, C.
European Journal of Combinatorics (2025).

[7] *Lascoux expansion of the product of a Lascoux and a stable Grothendieck* , with Orelowitz, G.
Combinatorial Theory (accepted); arXiv:2312.01647 .

[8] *Constructing maximal pipedreams of double Grothendieck polynomials* , with Chou, C.
The Electronic Journal of Combinatorics (2024).

[9] *Constructing a Gröbner basis of Griffin's ideal* .
Mathematische Zeitschrift (2024).

[10] *Connection between Schubert polynomials and top Lascoux polynomials* .
Algebraic Combinatorics (2025).

[11] *Top-degree components of Grothendieck and Lascoux polynomials* , with Pan, J.
Algebraic Combinatorics (2024).

[12] *A row analogue of Hecke column insertion* , with Huang, D. & Shimozono, M.
Combinatorial Theory (2024).

[13] *A bijection between K-Kohnert diagrams and reverse set-valued tableaux* , with Pan, J.
The Electronic Journal of Combinatorics (2023).

[14] *Set-valued tableaux rule for Lascoux polynomials* .
Combinatorial Theory (2023).

[15] *Grothendieck to Lascoux expansions* , with Shimozono, M.
Transactions of the American Mathematical Society (2023).

[16] *Harmonic bases for generalized coinvariant algebras* , with Rhoades, B. & Zhao, Z.
The Electronic Journal of Combinatorics (2020).

Selected Current Work

Two-row Schur times Schubert and parallel classes of positroid varieties

Gröbner geometry of hybrid pipe dreams, with Hamaker, Z. and Udell, G.

Sheared characters and forward-stable Lascoux polynomials, with Setiabrata, L.

Bumpless pipedreams for involutions and shifted plane partitions, with Marberg, E. and Morales, A. H.

Conference Proceedings and Extended Abstracts

[17] *Grothendieck positivity for square root crystals* , with Marberg, E. & Tong, K. H.
Proceedings of the 38th Conference on Formal Power Series and Algebraic Combinatorics; extended abstract of [2].

[18] *Grothendieck polynomials of fireworks and inverse fireworks permutations* , with Chou, J. C.-A. & Setiabrata, L.
Proceedings of the 38th Conference on Formal Power Series and Algebraic Combinatorics; extended abstract partially based on [6].

[19] *Generalized orthodontia, punched tableaux, and Lascoux expansion of sheared characters* , with St. Dizier, A. & Setiabrata, L.
Proceedings of the 38th Conference on Formal Power Series and Algebraic Combinatorics; extended abstract .

[20] *Marked bumpless pipedreams and compatible pairs* , with Huang, D. & Shimozono, M.
Proceedings of the 37th Conference on Formal Power Series and Algebraic Combinatorics; extended abstract of [4].

[21] *Analogues of two classical pipedream results on bumpless pipedreams* .
Proceedings of the 36th Conference on Formal Power Series and Algebraic Combinatorics; extended abstract of [5].

[22] *Grothendieck to Lascoux expansions* , with Shimozono, M.
Proceedings of the 34th Conference on Formal Power Series and Algebraic Combinatorics; extended abstract of [15].

[23] *Harmonic bases for generalized coinvariant algebras* , with Rhoades, B. & Zhao, Z.
Proceedings of the 33rd Conference on Formal Power Series and Algebraic Combinatorics; extended abstract of [16].

AWARDS AND GRANTS

University of California, San Diego

- Selma and Robert Silagi Award for Undergraduate Excellence (5000 USD)

TALKS

A positive combinatorial formula for the double Edelman–Greene coefficients

- University at Albany. Albany, NY. Sep. 2026
- University of Michigan. Ann Arbor, MI. Apr. 2026
- University of California, San Diego. La Jolla, CA. Feb. 2026
- University of California, Los Angeles. Los Angeles, CA. Feb. 2026
- Cornell University. Ithaca, NY. Jan. 2026
- Center for Combinatorics, Nankai University. Tianjin, China. Jan. 2026
- Sichuan University. Chengdu, China. Dec. 2025
- Beijing International Center for Mathematical Research. Beijing, China. Dec. 2025
- Canadian Mathematical Society Winter Meeting. Toronto, ON, Canada. Dec. 2025

An insertion algorithm for Schubert Cauchy identity via Pieri formula

- Geometry Seminar. Texas A&M University. College Station, TX. Mar. 2025
- Combinatorics Seminar. University of California, San Diego. La Jolla, CA. Mar. 2025

Grothendieck positivity for normal square root crystals

- Rutgers University. New Brunswick, NJ. Sep. 2026
- Formal Power Series and Algebraic Combinatorics conference . University of Washington. Seattle, WA. Jul. 2026
- Canadian Mathematical Society Summer Meeting. Université Laval. Québec, QC, Canada. Jun. 2025
- AMS Sectional Meeting. University of Kansas. Lawrence, KS. Mar. 2025
- Richard P. Stanley Seminar in Combinatorics. Harvard University. Cambridge, MA. Feb. 2025
- Combinatorics Seminar. University of Florida. Gainesville, FL. Jan. 2025
- Combinatorics Seminar. Laboratoire d'Algèbre, de Combinatoire et d'Informatique Mathématique. Montréal, QC, Canada. Dec. 2024

K-theoretic analogue of Demazure crystal

- Hong Kong University of Science and Technology. Hong Kong, China. Sep. 2024

Three combinatorial formulas of Lascoux polynomials

- Sun Yat-sen University. Guangzhou, China. Aug. 2024
- Guangdong Technion-Israel Institute of Technology. Shantou, China. Aug. 2024

Embedding bumpless pipedreams as Bruhat chains

- Formal Power Series and Algebraic Combinatorics conference . Bochum, Germany. Jul. 2024
- Center for Combinatorics, Nankai University. Tianjin, China. Jul. 2024
- Beijing International Center for Mathematical Research. Beijing, China. Jul. 2024
- Combinatorics and Representation Theory Seminar. Seoul, Korea (via Zoom). Apr. 2024
- Graduate Student Combinatorics Conference. Pittsburgh, PA. Mar. 2024
- The CAGE (Combinatorics, Algebra, and Geometry) Seminar. Philadelphia, PA. Mar. 2024
- Combinatorics Seminar. University of Southern California. Los Angeles, CA. Jan. 2024
- Combinatorial Algebra meets Algebraic Combinatorics. Montréal, QC, Canada. Jan. 2024
- Combinatorics Seminar. Laboratoire d'Algèbre, de Combinatoire et d'Informatique Mathématique. Montréal, QC, Canada. Dec. 2023

- Combinatorics Seminar. University of California, San Diego. La Jolla, CA. Nov. 2023
- Algebra Seminar. Virginia Polytechnic Institute and State University. Blacksburg, VA. Nov. 2023
- Combinatorics Seminar. University of Waterloo. Waterloo, ON, Canada. Sep. 2023
- Algebraic Combinatorics Seminar. Fields Institute. Toronto, ON, Canada. Sep. 2023

Pipedreams, bumpless pipedreams and Bruhat chains

- Schubert Seminar. Online. Nov. 2023

Top-degree components of Grothendieck and Lascoux polynomials

- Sichuan University. Chengdu, China (via Zoom). Nov. 2024
- Combinatorics Seminar. University of Michigan. Ann Arbor, MI. Mar. 2023
- Graduate Student Combinatorics Conference. St. Louis, MO. Mar. 2023
- Combinatorics Seminar. University of California, San Diego. La Jolla, CA. Jan. 2023
- Combinatorics Seminar. University of California, Los Angeles. Los Angeles, CA. Nov. 2022

A row analogue of Hecke column insertion

- Combinatorics Seminar. University of Florida. Gainesville, FL. Nov. 2022
- Combinatorics Seminar. University of Minnesota. Minneapolis, MN. Sep. 2022
- Combinatorics Seminar. University of Illinois Urbana-Champaign. Urbana, IL. Sep. 2022

Grothendieck to Lascoux expansions

- Combinatorics and Geometry Seminar. University of Washington. Seattle, WA. Apr. 2022
- Graduate Student Combinatorics Conference. San Diego, CA (via Zoom). Mar. 2022
- Algebra & Combinatorics Seminar. North Carolina State University. Raleigh, NC. Nov. 2021
- Algebra Seminar. Virginia Polytechnic Institute and State University. Blacksburg, VA. Nov. 2021
- Combinatorics Seminar. University of California, San Diego. La Jolla, CA. Oct. 2021
- Combinatorics Seminar. University of Waterloo. Waterloo, ON, Canada (via Zoom). Sep. 2021
- Combinatorics Seminar. University of Illinois Urbana-Champaign. Urbana, IL. Sep. 2021
- Combinatorics Seminar. University of Minnesota. Minneapolis, MN. Sep. 2021

Harmonic bases for generalized coinvariant algebras

- Formal Power Series and Algebraic Combinatorics conference . Ramat-Gan, Israel (via Zoom). Jan. 2022
- Combinatorics Seminar. University of California, San Diego. La Jolla, CA. Jun. 2021

TEACHING EXPERIENCE

Department of Mathematics, UCSD

Teaching Assistant: Calculus, Mathematical Reasoning, Combinatorics, Algebraic Combinatorics.

Department of Computer Science and Engineering, UCSD

Teaching Assistant: Discrete Mathematics, Design and Analysis of Algorithms, Theory of Computability.